

VCC Assignment 1 Report

1 Introduction

This report outlines the steps involved in setting up a virtualized environment using VirtualBox, configuring network settings for communication between multiple virtual machines (VMs), and deploying simple Node.js microservices on the VMs. The procedure includes creating two virtual machines, configuring network settings for communication, and testing microservice communication.

2 Creation and configuration of both VMs

Creation of Both VMs:

1. Using VirtualBox and clicking on **New** created both the VMs naming as **VMFirst** and **VMSecond**.
2. Allocated 2048 MB of RAM for each VMs.
3. Created a virtual hard disk (VDI with 25 GB) for both VMs.

Configuring both the VMs:

1. Selecting the VM and enabling the network adapter.
2. For Storage, added the Ubuntu ISO image to the virtual optical drive.

Installing the Guest OS:

1. Started the VMs and follow the Ubuntu installation prompts.
2. Once the installation is complete, installed VirtualBox Guest Additions.

3 Configuration of Network Settings to Connect the VMs

Setting Up a NAT Network:

1. In VirtualBox, added a new NAT Network.
2. Named the network `MicroservicesNetwork` and set the Network CIDR to `192.168.100.0/24`.

Configuring VMs Network Adapters:

1. For each VM, changed network types for both the Adapters 1 and 2.
2. For Adapter 1 to NAT Network and selected `MicroservicesNetwork`.
3. And for Adapter 2 to Host-only Adapter.

Testing Connectivity:

1. From each VM, ping the other VM's IP address using command:

```
ping 192.168.100.x
```

4 Deployment of a Simple Microservice Application

Install Node.js on Both VMs:

On both VMs, run the following commands:

```
sudo apt update
sudo apt install nodejs npm
```

Created Nodejs Microservice on both the VMs:

1. Created a directory for the microservice1 for VMFirst and microservice2 for VMSecond:

```
mkdir microservice1
mkdir microservice2
```

2. Initialized a Node.js project:

```
npm init -y
```

3. Installed Express:

```
npm install express
```

4. Created the `index.js` file with the code for nodejs microservice:

Starting the Microservices

1. On VM1, run:

```
node index.js
```

2. On VM2, run:

```
node index.js
```

Testing the Microservices

1. From VM1, used `curl` to test the service on VM2:

```
curl http://192.168.100.5:2500
```

2. From VM2, used `curl` to test the service on VM1:

```
curl http://192.168.100.4:3000
```

5 Conclusion

This procedure successfully sets up two VMs running Ubuntu, connects them via a NAT network, and deploys simple Node.js microservices on each VM. The microservices are able to communicate with each other across the network, demonstrating basic microservice architecture.

6 Architecture Design

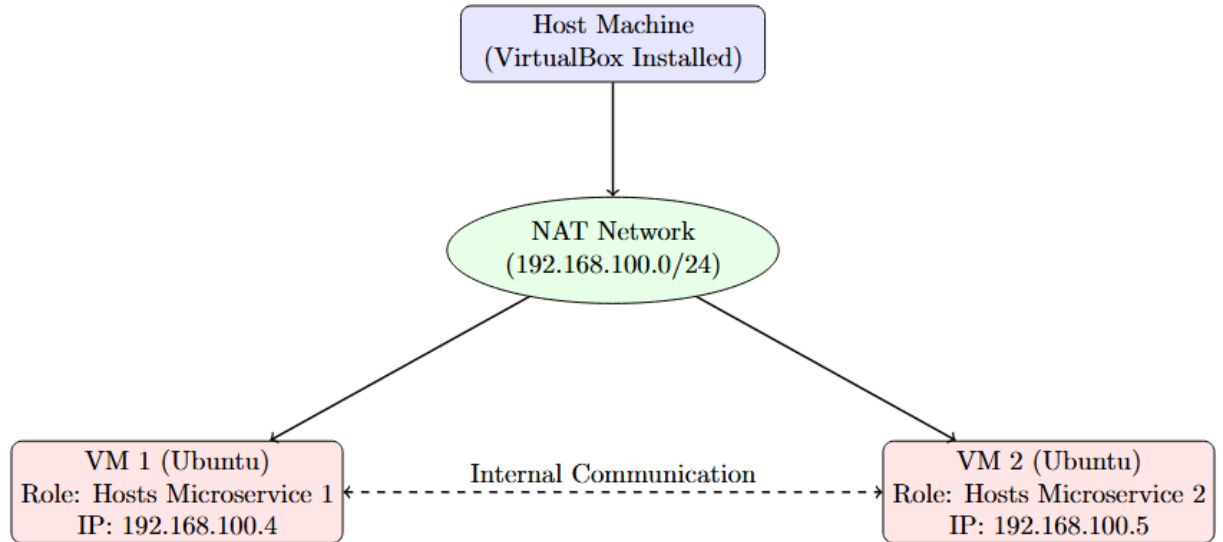


Figure 1: Architecture Design

7 Source Code Repo

Github Link

8 Recorded Video Demo

Video Demo